

CASE STUDY

A laptop comes back. What should happen next?

A Salesforce decision application for equipment recovery



A returned device might be sold again, repaired or recycled. ReUse Cloud compares eligible routes after costs and repair risk, explains its recommendation and separates evaluation from approval.

Native Apex + Lightning Web Components. Built by Ana Gomes. The cover is an explanatory illustration; the evidence pages contain real Salesforce captures and fictional portfolio data.

01 / THE PROBLEM AND THE SOLUTION

Choose a route you can explain

Eligibility comes before the financial comparison.

A device must have a condition score of at least 70 to be eligible for resale and at least 30 for repair. Recycling is always eligible. The engine compares unrounded expected contribution across eligible routes.

Expected contribution includes costs and uncertainty.

Acquisition credit and logistics make up the base cost. Resale subtracts that base from the sale price. Recycling subtracts it from salvage value. Repair accounts for probability-weighted proceeds and repair costs before subtracting the base.

Approval is a separate control.

A negative recommendation raises a manager-review flag. Only an Evaluated asset can be approved with the required permission. Changes to financial inputs reset the workflow to Draft, requiring reevaluation.

What the screen is for.

The workbench makes the route, expected contribution, review flag and current workflow state readable together. It supports inspection of the recommendation rather than presenting a score without an explanation.

02 / REAL PLATFORM CONTEXT

Three devices in the recovery workbench

The captured Salesforce screen shows three assets, two Evaluated, one Approved and one requiring manager review. The expected-contribution panel displays 150. The next pages enlarge the table and explain the calculations.

The screenshot displays the 'ReUse Cloud' interface within a Salesforce Developer Edition environment. The header includes the Salesforce logo, a search bar, and navigation icons. The main content area features a dark green banner with the text 'ReUse Cloud' and 'A second life. A defensible decision.' Below this, a summary section shows five metrics: VISIBLE ASSETS (3), EVALUATED (2), APPROVED (1), MANAGER REVIEW (1), and EXPECTED CONTRIBUTION (R\$ 150,00). The 'ReUse Cloud · Recovery Workbench' section includes instructions and buttons for 'Evaluate selected' and 'Refresh'. A table lists three assets with columns for Asset, Model, Condition, Route, Expected net, Review, Status, and an 'Inspect' button.

Asset	Model	Condition	Route	Expected net	Review	Status	Inspect
DEMO-001	Atlas 14	85	Repair	R\$ 620,00		Approved	Inspect
DEMO-002	Orion 15	65	Recycle	-R\$ 520,00	✓	Evaluated	Inspect
DEMO-003	Lumen 13	20	Recycle	R\$ 50,00		Evaluated	Inspect

The original full-resolution screen appears in the appendix. BRL is the demo display label; the values are modeled contributions, not money earned.

03 / READABLE EVIDENCE

The asset identities and route choices

The three existing records were inspected in the native Salesforce workbench. Atlas 14 recommends Repair; Orion 15 and Lumen 13 recommend Recycle.

☐	Asset	Model	Condition	Route
☐	DEMO-001	Atlas 14	85	Repair
☐	DEMO-002	Orion 15	65	Recycle
☐	DEMO-003	Lumen 13	20	Recycle

Original asset, model, condition and route columns, enlarged without changing their contents.

Expected net	Review	Status
R\$ 620,00		Approved
-R\$ 520,00	✓	Evaluated
R\$ 50,00		Evaluated

Original expected-net, review and status columns. Atlas: 620 and Approved. Orion: -520, review checked and Evaluated. Lumen: 50 and Evaluated.

04 / DECISION EXPLANATION

Why recycle Orion 15?

A positive resale figure is not enough.

Orion has condition 65. The resale calculation is 230, but resale is ineligible because the minimum condition is 70. The engine cannot choose an ineligible route just because its number is attractive.

Compare the options that are eligible.

Repair is eligible at condition 65, but its expected contribution is -807.50. Recycling contributes -520. Recycling is therefore the better eligible option, even though both have a negative result.

A negative result needs review.

The review flag makes that consequence explicit. The operator can see the recommendation without confusing it with an authorized approval. This case explains the existing inspected state; a new approval sequence was not recorded.

The other two records complete the picture.

Atlas 14 has condition 85 and recommends Repair at 620; its existing state is Approved. Lumen 13 has condition 20, making resale and repair ineligible; recycling contributes 50. Together, $620 - 520 + 50$ equals the panel total of 150.

A native engine behind the recommendation

The implementation is inspectable.

recoveryWorkbench calls RecoveryController. RecoveryEvaluationJob advances evaluation asynchronously. RecoveryAssetHandler and RecoveryDecisionEngine calculate and persist recommendations on Recovery_Asset__c.

Access and changed inputs matter.

Project permission sets, private sharing and user-mode queries support record access. Financial-input changes reset approval state so an old approval is not reused for a new calculation.

Validation is attached.

The recorded native deployment completed 15 Apex tests without failures. Decision engine, evaluation job and trigger reached 100% coverage in that run; other classes have partial coverage. Source and test evidence are included separately.

Evidence boundaries.

The workbench and existing record outcomes were inspected in Salesforce. The video uses authentic captures with explanatory animation. It does not pretend to record every approval click or claim a real client deployment.

Appendix A / Original Salesforce workbench capture

Developer Edition

Pesquisar...

Developer Edition

Welcome

ANA GOMES · SALESFORCE / APEX / LWC

ReUse Cloud

A second life. A defensible decision.

Compare resale, repair and recycling contribution before approving equipment recovery.

Developer Edition · Fictional portfolio lab · BRL

VISIBLE ASSETS

3

EVALUATED

2

APPROVED

1

MANAGER REVIEW

1

EXPECTED CONTRIBUTION

R\$ 150,00

ReUse Cloud · Recovery Workbench

Compare expected contribution, inspect the recommendation, then approve unchanged inputs. Demo currency: BRL.

Evaluate selected

Refresh

Most recent 200 visible assets. Recommendations are calculated on save; evaluation advances the workflow asynchronously.

☐	Asset	Model	Condition	Route	Expected net	Review	Status	
☐	DEMO-001	Atlas 14	85	Repair	R\$ 620,00		Approved	Inspect
☐	DEMO-002	Orion 15	65	Recycle	-R\$ 520,00	✓	Evaluated	Inspect
☐	DEMO-003	Lumen 13	20	Recycle	R\$ 50,00		Evaluated	Inspect

Appendix B / Native deployment and 15 Apex tests

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Developer Edition

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Configuração de pesquisa

★

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🔊

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⚙️

🔔2

👤

⚙️

Configuração

Início

Gerenciador de objetos

🔍 Busca rápida

Página inicial da Configuração

Salesforce Go

Relatórios da Service Cloud

Assistente de configuração do Commerce

Assistente do Hyperforce

Atualizações da versão

Aplicativo móvel do Salesforce

Uso do Lightning

Ofertas do AgentExchange

Sales Cloud Everywhere

ADMINISTRAÇÃO

> Usuários

> Dados

> Email

FERRAMENTAS DA PLATAFORMA

> Gerenciamento de assinatura

> Aplicativos

> Configurações de recurso

> Slack

> Data Cloud

> Heroku

> MuleSoft

⚙️

CONFIGURAÇÃO

Status da implantação

Detalhes da implantação

[Ajuda para esta página](#)

[« Voltar para o Status da implantação](#)

Implantação bem-sucedida

Nome: 0Afg700000EES0T

Tipo: API

Implantado por: [Ana Gomes](#)

Hora de início: 04/10/2026 01:02

Hora de término: 04/10/2026 01:02

Número de arquivos ⓘ : 29

Tamanho total descompactado ⓘ : 44.930 bytes

1

Implantar componentes

38/38

2

Executar testes do Apex

15/15

Original capture at 1 image pixel per PDF point. Zoom or open this page at actual size. Platform content unchanged.